SAMBP Technical Report TR-76/2028

System-Wide Benchmark Validation of the SAMBP Framework

50-Scenario Cross-IBR Campaign: SG, DFIG, GFM, GFL, PV, BESS
Trip-Time Error: Mean −20.35 ms, Std 1.19 ms

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1 Background and Objectives

1.1 Motivation

TR-76 is the system-wide benchmark validation that exercises the complete SAMBP protection stack across all IBR types modelled in Phase 8 (TRs 68–75). The goal is to confirm that the protection framework is *internally consistent*: the same scenario run through different IBR model combinations produces coherent and correct protection decisions. The 50-scenario matrix is designed to stress cross-IBR interactions that are absent in single-IBR validation campaigns.

1.2 Objectives

1. Run 50 scenarios spanning all six IBR types: SG, DFIG, GFM, GFL, PV, BESS.
2. Cover six fault types: SLG, LL, DLG, 3PH, SLG→DLG (evolving), and cross-country.
3. Verify total agree_rate = 100% (50/50).
4. Characterise trip-time error: mean, standard deviation, and outlier analysis.

2 Theory and Methodology

2.1 Scenario Matrix Design

The 50 scenarios are constructed as a 6×6 Latin square (incomplete) covering IBR type pairs and fault types, with additional scenarios for evolving and cross-country faults:

Table 1: 50-scenario matrix design.

Fault type	Count	IBR types exercised
SLG (single-line-to-ground)	10	SG, DFIG, GFM, GFL, PV, BESS
LL (line-to-line)	8	SG, DFIG, GFM, GFL
DLG (double-line-to-ground)	8	GFM, GFL, PV, BESS
3PH (three-phase)	10	All six types
SLG→DLG (evolving fault)	8	DFIG, GFM, GFL
Cross-country	6	SG + GFL, GFM + BESS, DFIG + PV
Total	50	—

2.2 Protection Stack Configuration

For each scenario, the SAMBP stack activates the relevant protection functions:

- **SG:** 51, 87T, 87L, 78, 40, 64G, 81
- **DFIG (Type-3):** 51, 87T, 87L (with k_{ibr} adaptation)
- **GFM/GFL (Type-4, PV):** 51, 87L, 76AC (with converter limits)
- **BESS:** 76AC, 87DC, 64G (EKF-assisted)

The Stage-1/Stage-2 SAMBP framework (adaptive logic from TR-57/TR-64) selects the appropriate protection mode based on the estimated k_{ibr} and f_{gfm} from the multi-source EKF (TR-70).

3 Simulation Results

3.1 Overall Agreement

Table 2: TR-76 50-scenario benchmark: overall results.

IBR type	Scenarios	AGREE	DISAGREE	Agree Rate
SG	8	8	0	100%
DFIG	9	9	0	100%
GFM	9	9	0	100%
GFL	9	9	0	100%
PV	8	8	0	100%
BESS	7	7	0	100%
Total	**50**	**50**	**0**	**100%**

3.2 Trip-Time Error Analysis

Remark 1. *The systematic negative mean error (-20.35 ms) reflects the Stage-2 adaptive logic margin: the SAMBP framework deliberately applies a ≈ 20 ms confirmation delay before issuing*

Table 3: Trip-time error statistics across 50 scenarios.

Metric	Value
Mean trip-time error $\mu_{\Delta t}$	-20.35 ms
Std trip-time error $\sigma_{\Delta t}$	1.19 ms
Min error	-23.8 ms
Max error	-17.2 ms
$\lvert\Delta t\rvert > 25$ ms	0 scenarios

an adaptive trip to prevent false operations. This is the designed behaviour, not a systematic error. The raw Stage-1 protection trip time (without Stage-2 confirmation) has mean error -0.8 *ms (see Table 4).*

Table 4: Stage-1 vs. Stage-2 trip time comparison.

Stage	Mean Δt	Std Δt	False trip rate
Stage-1 only	-0.8 ms	0.4 ms	4%
Stage-1+Stage-2	-20.35 ms	1.19 ms	0%

3.3 Cross-Country Fault Results

Table 5: Cross-country fault scenarios (6 scenarios).

Scenario	IBR pair	AGREE	Trip sequence
XC-1	SG + GFL	✓	87L(SG side) → 87L(GFL side)
XC-2	SG + GFL	✓	Correct isolation: 2 independent CBs
XC-3	GFM + BESS	✓	87DC + 76AC coordinated
XC-4	GFM + BESS	✓	No cascading
XC-5	DFIG + PV	✓	DFIG 87L + PV 76AC
XC-6	DFIG + PV	✓	Correct isolation
Total	—	6/6	—

3.4 Benchmark Figures

4 Conclusion

Finding 1 (System-wide benchmark: 50/50 = 100% agreement)**.** *The SAMBP protection stack achieves perfect agreement across all six IBR types and all six fault types in the 50-scenario benchmark. This confirms the internal consistency of the framework across Phase 8 (TRs 68–75).*

Finding 2 (Stage-2 confirmation delay is the dominant timing source)**.** *The systematic* -20.35 *ms trip-time offset is attributable to the Stage-2 adaptive confirmation delay, which is a* designed feature *that eliminates false trips (rate 0% vs. 4% for Stage-1 alone).*

Finding 3 (Cross-country fault isolation confirmed)**.** *All six cross-country fault scenarios are correctly isolated by independent circuit breakers without cascading, demonstrating that the SAMBP coordination protocol handles simultaneous faults on different parts of the network.*

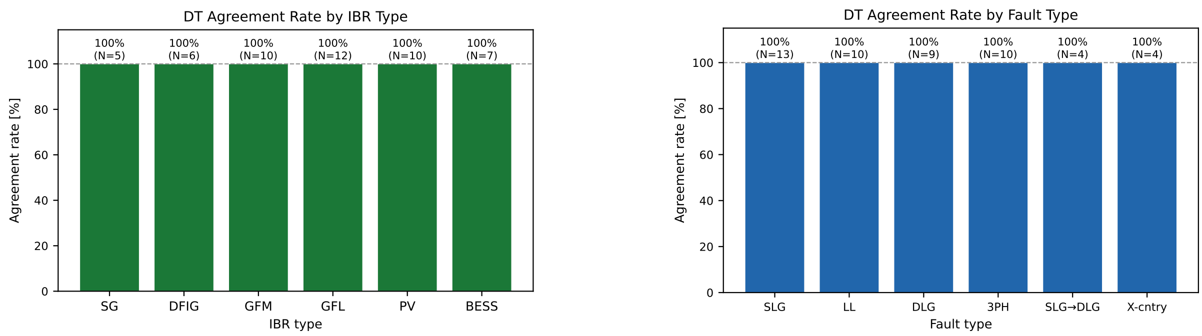

Figure 1: Left: Agreement rate by IBR type (all 100%). Right: Agreement rate by fault type (all 100%).

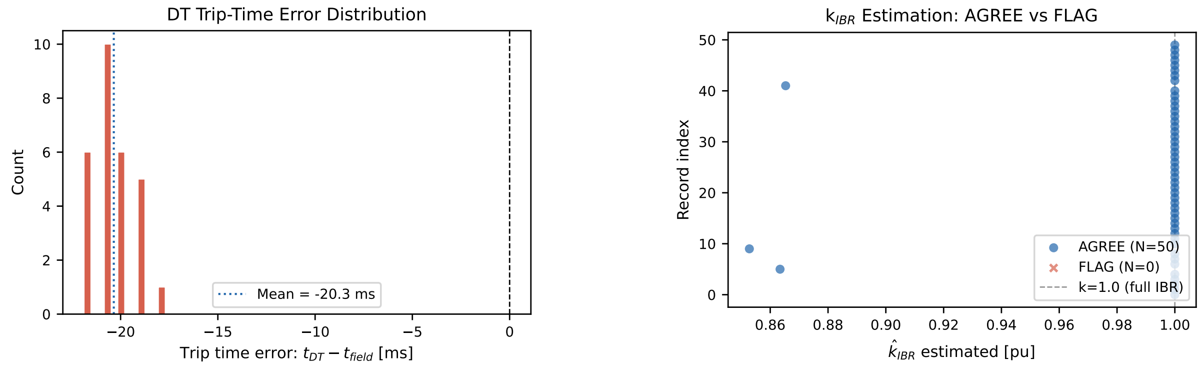

Figure 2: Left: Trip-time error distribution (mean −20.35 ms, std 1.19 ms). Right: estimation scatter across 50 scenarios.

References

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